

MID VALLEY, TX, USA

WMO: 722159

Lat: 26.178N

Lon: 97.973W

Elev: 21

StdP: 101.07

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12980

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.6	4.9	-8.6	1.8	17.8	-5.3	2.4	16.9	11.9	24.3	10.9	24.1	3.8	350	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	37.6	24.1	36.9	24.3	36.0	24.4	27.2	30.7	26.8	30.7	26.4	30.6	5.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.4	21.9	27.9	26.1	21.6	27.9	25.6	20.9	27.9	85.4	30.1	83.6	29.8	82.3	30.5	33.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.6	9.2	8.4	DB	0.4	39.5	1.8	1.0	-0.9	40.2	-1.9	40.7	-2.9	41.3	-4.3	42.0
			WB	-1.2	27.9	1.8	1.8	-2.5	29.2	-3.5	30.2	-4.5	31.2	-5.8	32.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.2	15.9	18.6	21.6	24.6	27.6	29.8	29.9	30.8	28.4	25.2	20.7	17.0
	DBStd	6.30	5.26	5.49	4.41	3.27	2.56	1.46	1.57	1.37	2.03	3.60	4.69	5.29
	HDD10.0	29	13	6	1	0	0	0	0	0	0	0	2	7
	HDD18.3	321	108	60	23	3	0	0	0	0	0	3	31	92
	CDD10.0	5214	197	247	361	438	546	593	617	645	553	471	323	223
	CDD18.3	2463	34	67	124	191	287	343	359	387	303	216	101	50
	CDH23.3	26522	208	436	859	1665	3056	4242	4505	5032	3280	2139	787	313
Wind	WSAvg	3.8	3.6	4.2	4.4	4.5	4.5	4.1	4.2	3.7	2.7	3.0	3.3	3.5
	PrecAvg	632	24	26	29	42	58	58	60	51	136	81	36	30
	PrecMax	814	82	86	162	166	157	182	279	196	273	213	130	124
	PrecMin	405	3	0	2	1	1	2	0	0	31	1	2	2
	PrecStd	143	23	25	36	47	44	47	81	46	61	58	32	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.4	32.4	33.8	35.2	36.3	37.9	38.5	38.6	37.7	35.0	32.4	30.3
		MCWB	19.8	19.3	21.8	23.6	24.7	23.7	23.9	23.7	23.8	23.9	23.1	21.4
	2%	DB	27.5	29.9	31.2	32.8	35.0	36.6	37.2	37.7	36.1	33.7	31.0	28.4
		MCWB	20.2	20.7	21.7	23.4	24.8	24.3	24.4	24.1	24.1	24.4	22.7	21.1
	5%	DB	26.0	27.9	29.6	32.0	33.8	35.8	36.1	37.1	34.8	32.6	29.3	27.0
		MCWB	18.9	20.6	21.5	22.7	24.4	24.2	24.4	24.4	24.4	23.8	21.7	20.5
	10%	DB	23.8	26.2	27.9	30.5	32.7	34.8	34.8	36.0	33.3	31.4	27.8	25.0
		MCWB	18.4	19.7	20.6	22.2	24.2	24.1	24.1	24.6	24.7	23.4	21.2	20.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	23.3	24.1	25.8	27.2	27.7	27.5	27.3	27.5	27.4	25.1	23.6
		MCDB	25.8	27.9	30.0	30.9	31.8	30.5	30.5	31.1	31.1	30.0	29.6	27.5
	2%	WB	21.7	22.5	23.2	25.1	26.3	26.9	26.9	26.8	26.8	26.4	24.1	22.8
		MCDB	25.0	26.6	27.8	30.0	30.8	30.7	30.5	31.0	30.7	29.9	28.3	26.1
	5%	WB	20.8	21.9	22.7	24.4	25.8	26.5	26.5	26.5	26.3	25.7	23.4	22.0
		MCDB	24.0	25.5	26.8	28.9	30.3	30.8	30.4	30.9	30.4	29.6	26.9	25.3
	10%	WB	19.8	21.2	22.1	23.7	25.3	26.0	26.1	26.1	25.8	25.1	22.7	21.1
		MCDB	22.9	24.4	25.7	28.0	29.9	30.6	30.3	30.8	30.1	29.2	26.0	24.0
Mean Daily Temperature Range	5% DB	MDBR	10.9	10.9	10.5	10.8	9.9	10.3	10.1	10.9	9.9	11.2	11.1	10.5
		MCDBR	12.0	11.8	11.2	11.5	10.4	11.3	11.2	11.7	11.7	11.2	11.3	11.7
		MCWBR	5.9	5.0	4.0	4.3	2.9	2.4	2.4	2.4	2.9	3.6	4.3	5.9
	5% WB	MCDBR	10.4	10.0	9.5	10.0	9.2	9.5	9.3	9.9	9.1	9.4	10.1	10.0
		MCWBR	6.2	5.0	4.1	4.0	3.1	2.7	2.7	2.6	2.9	4.0	5.0	6.1
Clear-Sky Solar Irradiance	taub		0.352	0.369	0.394	0.425	0.431	0.416	0.431	0.421	0.427	0.382	0.370	0.354
	taud		2.479	2.422	2.352	2.279	2.298	2.395	2.359	2.396	2.346	2.466	2.451	2.479
	Ebn at Noon		899	908	901	877	866	873	859	869	857	883	871	881
	Edh at Noon		96	109	124	136	134	121	125	120	123	103	98	93
All-Sky Solar Radiation	RadAvg		3.15	3.82	4.67	5.52	6.03	6.67	6.63	6.35	5.26	4.59	3.55	2.85
	RadStd		0.35	0.57	0.53	0.49	0.41	0.24	0.46	0.28	0.39	0.44	0.38	0.40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105

Nomenclature: See separate page